Joel Biswas

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https://github.com/JoelBiswas

Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Electrical Engineering; GPA: 4.00/4.00 *Exp. May 2028*

Experience

Digital Design and Verification Member September 2026 – Present
Georgia Tech SiliconJackets *Atlanta, GA*

- Worked across RTL design, functional verification, synthesis, and physical design as a member of Georgia Tech's student ASIC design team.
- Developed a constrained-random SystemVerilog verification environment with assertions, monitors, and a reference-model scoreboard to detect and isolate processor corner-case bugs.
- Executed SKY130 physical-design workflows in Cadence Innovus, including SRAM macro floorplanning, power-ring generation, placement, and setup timing analysis.

Electrical Engineering Intern June 2025 – June 2026
Origami Robotics LLC *Suwanee, GA*

- Designed a custom 4-layer ESP32-C3 PCB integrating CAN, dual I^2C GPIO expanders, USB-C, onboard power regulation, and 30 GPIO channels.
- Programmed and tested embedded hardware interfaces, validating peripheral communication, power delivery, and controller functionality during system bring-up.
- Designed power, signal-integrity, and protection circuitry including 3.3 V buck regulation, IC decoupling, USB-C TVS protection, and differential routing for USB and CAN.

Team Captain & Electrical Lead May 2025 – May 2026
Lambert Robotics FRC Team 9477 *Suwanee, GA*

- Led a ∼60-member engineering team across electrical, programming, CAD, manufacturing, and system integration in the development of competition robots.
- Designed and debugged robot-wide electrical systems integrating CAN networks, sensors, motor controllers, power distribution, and embedded control hardware.

Projects

32-bit RISC-V Processor | *SystemVerilog, Cadence Xcelium, Cadence Innovus* September 2026

- Designed a 32-bit RISC-V CPU in SystemVerilog with a single-cycle datapath, multi-cycle SRAM loads, and support for arithmetic, shifts, branches, and load/store instructions.
- Built and verified processor components including instruction decoding, register file, ALU, control logic, and instruction/data SRAM interfaces using directed and randomized tests.
- Synthesized the CPU targeting SKY130, integrating two 1,024 × 32-bit SRAM macros and 8,400 reported cells; performed floorplanning, placement, power planning, and static timing analysis in Cadence Innovus.

Custom R2-D2 Robot | *ESP32, C++, nRF24L01+, CAD* July 2025

- Designed and built a mobile three-wheel robotic platform integrating DC gearmotors, motor drivers, servos, power regulation, and stepper-driven dome rotation.
- Developed wireless embedded control using microcontrollers and nRF24L01+ transceivers, implementing joystick-based drive control and remote actuation.
- Iteratively redesigned the drivetrain after identifying insufficient motor torque in the initial prototype, recalculating mechanical requirements and selecting higher-torque geared motors to achieve reliable locomotion.

Technical Skills

HDL/Programming: SystemVerilog, Verilog, C++, Python, C
Digital Design: RTL Design, RISC-V, Computer Architecture, Digital Logic, Static Timing Analysis, Synthesis, Physical Design
Verification: Constrained-Random Verification, Assertions, Scoreboards, Testbenches, Functional Verification
Embedded Systems/Hardware: PCB Design, Embedded Systems, ESP32, CAN, SPI, I^2C, UART, PWM
Tools: Cadence Xcelium, Cadence Innovus, Synopsys Design Compiler, KiCad, Git, Linux, Fusion 360, Onshape